

The $SU(2)$ mid-plane on its small-field set: bounds and the missing comparison

After refereeing (GPT-6 Astra, 2026-09-28). Claim-by-claim review of Claude's derivation: **A1 ACCEPT** (with the inverse notation corrected); **A2 REFINE** (complete image pairing, explicit constants, endpoint range, and the distinction between a bridge and the interacting measure); **A3 REFINE** (local convexity survives with corrected midpoint curvature and defined derivative constants); **A4 REJECT** (Proposition 4 is false under Hypothesis I, already along a flat-holonomy path; its covariance, gauge comparison and Taylor estimates also have missing premises); **A5 REFINE** (the Peierls arithmetic survives with corrected curvature constant; the logarithmic size condition remains conditional). The accepted results are Lemmas 1–2 and the corrected local Lemma 3 below. The normalized small-field comparison remains open. The original proof and its attribution are preserved in git history.

Take one halving of direction 1 in $1 + 2$ dimensions, with gauge group $SU(2)$, heat time $t = \lambda_3 a = \hbar g_{\text{cl}}^2 a$, and

$$\varepsilon = t^{1/2-\delta}, \quad \eta = 2\varepsilon, \quad 0 < \delta < 1/6.$$

All operator norms below use the Euclidean norm on the three real colour components, $[T_a, T_b] = \epsilon_{abc} T_c$, $|T_a| = 1$. Distances on $SU(2)$ use the three-sphere of radius 2; Haar measure has total mass one. Inputs read at passage level: series/parallel, Proposition 1, Hypothesis P(α), Proposition 7; mid-plane order t , §§1 and 7, especially (16)–(18); exact midpoint, Theorem 1 and its image formula. This note concerns one finite mid-plane and bounds uniform in its area where explicitly stated.

1. Setting and the restriction

There are N sites, $2N$ edges and N faces on the periodic mid-plane. Use periods at least three to avoid repeated boundary incidences. In the mid-vertex gauge of Proposition 1 set $m_e = m_{*e} \exp \xi_e$, with m_{*e} the shortest-geodesic midpoint between P_e and Q_e^{-1} . Their distance is $|X_e| \leq \varepsilon$. Up to constants independent of ξ , the exponent is

$$S(\xi; U) = \frac{1}{t} \sum_e \left[d(P_e, m_e)^2 + d(m_e, Q_e^{-1})^2 - \frac{1}{2} |X_e|^2 \right] + \frac{1}{4t} \sum_g |\log m_{\partial g}|^2 + J(\xi; U). \quad (1)$$

Here J contains the full heat-kernel and Haar amplitudes. Its local ξ derivatives are bounded uniformly on a fixed chart away from the cut locus. Constants in J , including its bridge value at $\xi = 0$, can be transferred to the external normalization; only the bridge's squared-distance term necessarily vanishes there. Count each bridge denominator once, either in S or outside the integral. Include the old-face ratios in the external normalization $B(U)$.

Keep Claude's soft-barrier definition explicit:

$$W_\eta(\xi) = \sum_e w_\eta(|\xi_e|), \quad \mu_U(d\xi) = Z_s(U)^{-1} e^{-S(\xi; U) - W_\eta(\xi)} d\xi, \quad \mathcal{D}_s = -\log Z_s + B(U). \quad (2)$$

The domain is $\prod_e B(0, c_0)$; w_η is C^2 , convex and radial, vanishes for $r \leq \eta$, and diverges at c_0 . Choose its boundary decay so that the integrations by parts used below have vanishing boundary terms and finite Fisher information. It is fixed independently of U in these midpoint coordinates. Its support extends beyond $\Omega_\eta = \{\max_e |\xi_e| \leq \eta\}$. A hard restriction to Ω_η defines a different partition function. Both choices are convex restrictions, but their covariance lower bounds need different boundary arguments.

The interacting law includes every mid-face weight. Lemma 2 bounds the single-edge bridge before these weights are inserted. Transferring that tail bound to (2), bounding the probability that any of $2N$ edges is large, and comparing the two partition functions are additional tasks. Even for independent bridges the union bound carries a factor $2N$.

2. Lemma 1: covariant inverse decay (A1 ACCEPT)

Let C_A be the oriented face-edge incidence operator with each entry an adjoint rotation of the background connection. Set $H_A = 4I + \frac{1}{2}C_A^*C_A$. For every background,

$$4I \leq H_A \leq 8I. \quad (3)$$

If $K = H_A + E$, $E = E^*$ has range one in the graph of edges sharing a face, and $\|E\| \leq \kappa < 4$, then, writing $d = d(e, e')$,

$$\|(K^{-1})_{ee'}\| \leq \frac{1}{4-\kappa} \left(\frac{2+\kappa}{6} \right)^d. \quad (4)$$

In particular the original range $\kappa < 2$ is valid, and at $\kappa = 0$ the bound is $\frac{1}{4}3^{-d}$. Here K denotes the perturbed Hessian; K^{-1} denotes its inverse.

Proof. For each face, $|(C_A\xi)_g|^2 \leq 4 \sum_{e \in \partial g} |\xi_e|^2$. Each edge belongs to two faces, so $\|C_A\xi\|^2 \leq 8\|\xi\|^2$. Thus $\|K - 6I\| \leq 2 + \kappa < 6$. The n th power of $K - 6I$ has zero (e, e') block for $n < d$, and each remaining block has norm at most $(2 + \kappa)^n$. Sum the Neumann series about $6I$ to obtain (4). The count of faces per edge makes every constant independent of N . $\square$

This is a bound on the sum of operator products at a given order. It supplies no bound of 3^{-L} for each individual path with the same constant after absolute path counting.

3. Lemma 2: heat kernel and bridge tail (A2 REFINE)

For $0 < t \leq 1$ and $0 \leq \theta \leq 2\pi$,

$$k_t(\theta) \leq C_+ t^{-5/2} e^{-\theta^2/(2t)}; \quad k_t(\theta) \geq c_- t^{-3/2} e^{-\theta^2/(2t)} \quad (0 \leq \theta \leq 1). \quad (5)$$

Here are explicit, deliberately loose constants. Put $a_n = 4\pi n$, $b_n = (4n + 2)\pi$, and

$$\begin{aligned} A_0 &= \pi \left[1 + 2 \sum_{n \geq 1} (1 + (a_n + \pi)^2) e^{-(a_n^2 - 2\pi a_n)/2} \right], \\ A_1 &= 2\pi \sum_{n \geq 0} (1 + (b_n + \pi)^2) e^{-b_n(b_n - 2\pi)/2}, \\ C_+ &= e^{1/8} \sqrt{8\pi} \max(A_0, A_1), \quad c_- = \sqrt{8\pi}. \end{aligned} \quad (6)$$

Written image check. Write $f_t(u) = u e^{-u^2/(2t)}$. The exact image formula is

$$k_t(\theta) = e^{t/8} \sqrt{8\pi} t^{-3/2} \frac{\sum_{w \in \mathbb{Z}} f_t(\theta - 4\pi w)}{\sin(\theta/2)}.$$

On $[0, \pi]$, pair $w = \pm n$. Their sum is $f_t(a_n + \theta) - f_t(a_n - \theta)$, which vanishes at zero. Since $|f'_t(u)| \leq (1 + u^2/t) e^{-u^2/(2t)}$, its absolute value is at most

$$2\theta(1 + (a_n + \pi)^2/t) e^{-(a_n - \theta)^2/(2t)}.$$

Use $\theta/\sin(\theta/2) \leq \pi$ and $a_n^2 - 2a_n\theta \geq a_n^2 - 2\pi a_n$ to obtain the A_0/t bound on the image quotient after extracting $e^{-\theta^2/(2t)}$.

On $[\pi, 2\pi]$, set $\psi = 2\pi - \theta \in [0, \pi]$. Pair **all** images as $w = -n$ and $w = n + 1$, $n \geq 0$. Each pair is $f_t(b_n - \psi) - f_t(b_n + \psi)$ and has magnitude at most

$$2\psi(1 + (b_n + \pi)^2/t)e^{-(b_n - \psi)^2/(2t)}.$$

Divide by $\sin(\psi/2) \geq \psi/\pi$ and use $(b_n - \psi)^2 - (2\pi - \psi)^2 \geq b_n(b_n - 2\pi)$. This gives A_1/t , including the antipodal limit $\psi = 0$. In particular the leading pair is

$$e^{-(4\pi^2 + \psi^2)/(2t)}[(2\pi - \psi)e^{2\pi\psi/t} - (2\pi + \psi)e^{-2\pi\psi/t}],$$

as in Claude's calculation. Pairing the remaining images too removes the apparent singularity from division by $\sin(\psi/2)$.

For $\theta \leq 1$, the absolute remainder relative to $\theta e^{-\theta^2/(2t)}$ is bounded by

$$2 \sum_{n \geq 1} (1 + (a_n + 1)^2/t) e^{-(a_n^2 - 2a_n)/(2t)} < \frac{1}{2}.$$

For completeness, $3 < \pi < 4$ bounds this sum by $2 \sum_{n \geq 1} (1 + 289n^2) e^{-56n^2} < 1/2$; use $t^{-1} e^{-b/t} \leq e^{-b}$ for $b \geq 1$, $t \leq 1$, followed by $n^2 \leq 4^n$, $e^{56} > 2^{56}$ and a geometric series. Together with $\theta/\sin(\theta/2) \geq 2$ this proves the lower bound with (6), also at zero by continuity. $\square$

Bridge conclusion. Assume $|X| \leq 1$, $r \geq |X|/2$, and $0 < t \leq 1$. For the normalized bridge of duration t ,

$$\beta\{d(m, m_*) \geq r\} \leq \min\left(1, C_2 t^{-7/2} e^{-2r(r-|X|)/t}\right), \quad C_2 = 32C_+^2/c_- . \quad (7)$$

Indeed at distance $\rho \geq r$ both endpoint distances are at least $\rho - |X|/2 \geq 0$. The two upper bounds at time $t/2$ divided by the lower bound at time t give the density bound with exponent $-[2(\rho - |X|/2)^2 - |X|^2/2]/t = -2\rho(\rho - |X|)/t$. This exponent decreases with ρ for $\rho \geq |X|/2$; integrate against Haar measure of mass one. At $r = \eta = 2\varepsilon$ and $|X| \leq \varepsilon \leq 1$ the gain is $e^{-4\varepsilon^2/t}$. This proof uses the two endpoint triangle inequalities through m_* .

4. Lemma 3: a corrected local convexity theorem (A3 REFINE)

The midpoint background face logarithm $Y_g = \log(m_* \partial g)$ obeys $|Y_g| \leq 3\varepsilon$. To see this, compare its four edges to those of the transported lower layer. Each midpoint changes its edge by distance at most $\varepsilon/2$, and the lower-layer plaquette has distance at most ε . Bi-invariance and the product triangle inequality give $\varepsilon + 4\varepsilon/2 = 3\varepsilon$. The bound ε in the original sketch needs an extra premise. Individual background links can be large; transport them out of each local product before taking logarithms.

Here is a precise version of the local estimate, with constants defined by fixed-dimensional derivatives rather than unquantified BCH symbols. Fix $r_0 = 1/16$. On $|b|, |z| \leq r_0$ define

$$B(b, z) = d(e, e^{-b/2} e^z)^2 + d(e, e^{b/2} e^z)^2 - |b|^2/2,$$

and on $|y|, |z_i| \leq r_0$ define

$$F(y, z_1, \dots, z_4) = \frac{1}{4} |\log(e^y e^{z_1} \dots e^{z_4})|^2.$$

All these products lie in a fixed injectivity chart. Let L_B be the supremum of the derivative of $D_z^2 B$ with respect to (b, z) , using the sum norm on the input blocks and operator norm on the output. Let L_F be the analogous supremum of the derivative of $D_{(z_1, \dots, z_4)}^2 F$. These are finite constants on the indicated compact sets. At the origin these Hessians are $4I$ and the block matrix with every block $I/2$, respectively.

Let J_B, J_F be supremum norms of the edge and face Hessians of the corresponding local heat-kernel/Haar log amplitudes for $0 < t \leq 1/2$ on the same charts; omit terms independent of ξ . They are finite: the zero-image Jacobian factors are analytic on these compact sets, and

each differentiated nonzero image has a polynomial in $1/t$ times $e^{-c/t}$ with $c > 0$. The positive zero-image quotient bounds the logarithm's denominator away from zero for small t ; positivity and compactness handle the remaining closed time interval. Define

$$J_* = J_B + 2J_F, \quad C_3 = L_B + 8L_F + J_*. \quad (8)$$

These supremum definitions fix the constants independently of the plane; a numerical value is unnecessary for the following explicit condition. For $3\varepsilon \leq r_0$, $\rho = |\xi|_\infty \leq r_0$, $0 < t \leq \min(1/2, \varepsilon)$, the **unbarriered** action satisfies

$$\|t \nabla_\xi^2 S - H_A\| \leq L_B(\varepsilon + \rho) + 2L_F(3\varepsilon + 4\rho) + tJ_* \leq C_3(\varepsilon + \rho). \quad (9)$$

Here C_A uses the signed adjoint transports in the ordered background face products. At each face the four rotations are isometries. Applying the mean value theorem to the two Hessians just defined gives the local bounds; summing face quadratic forms costs a factor two because every edge belongs to two faces. This proves (9) directly and supplies the derivative control missing from the BCH sketch.

Choose $c_0 \leq r_0$ with $C_3(\varepsilon + c_0) < 4$, and $2\varepsilon < c_0$. Then S is uniformly convex on the product of balls of radius c_0 ; $S + W_\eta$ has the same lower Hessian bound. On Ω_η the unbarriered error is at most $3C_3\varepsilon$. The soft measure (2) has the weaker pointwise error $C_3(\varepsilon + c_0)$ on its full support. Its barrier Hessian can be arbitrarily large. These distinctions matter in §5.

5. Proposition 4: verdicts and the corrected conditional implication

The original Hypothesis I asked for a C^2 path $U(s)$ from 1 to U with $|X_p(s)| \leq 2\varepsilon$,

$$\sum_p |\dot{X}_p(s)|^2 \leq C_I \sum_p |X_p(1)|^2, \quad |\ddot{X}_p(s)| \leq C_I \sum_{p' \sim p} |X_{p'}(1)|^2. \quad (10)$$

A4 overall: REJECT, with a counterexample. On a fixed periodic plane choose identical adjacent flat layers with commuting holonomies $e^{s\theta_2 T_3}, e^{s\theta_3 T_3}$. Constant links $e^{s\theta_j T_3/N_j}$ realize the path. Every plaquette flux, flux velocity and flux acceleration is zero, so (10) holds. Nevertheless, the fixed-volume Laplace calculation in (16)–(17) of the mid-plane note gives

$$\mathcal{D}_s(U_\theta, t) - \mathcal{D}_s(1, t) = \Omega_N(\theta) + o_N(1), \quad \Omega_N(\theta) > 0 \quad \text{for a nontrivial adjoint holonomy.} \quad (11)$$

The same leading term holds for (2): the barrier equals zero in a ball of radius η , and $\eta/\sqrt{t} = 2t^{-\delta} \rightarrow \infty$; rescaling by $\sqrt{t}$ recovers the full Gaussian determinant at fixed N . Uniform convexity about this zero-flux saddle controls the discarded tail at fixed N . The proposed right-hand side $C\varepsilon \sum_p |X_p|^2/t$ is exactly zero. Hypothesis I therefore allows data contradicting the claimed theorem. A winding term or an additive volume remainder must be included, with its own estimate. The actual $P(\alpha)$ allows $Ct^\alpha \sum_p 1$ and a size clause; those features distinguish it from the rejected stronger assertion.

A4(a), linear term: ACCEPT in a smooth local chart at 1. Global conjugation rotates every Lie-algebra link tangent by the adjoint representation. Its invariant covectors vanish, so the first derivative of the gauge-invariant partition function at 1 is zero. The same argument works for independent flux coordinates if such a chart has been supplied. Plaquette logarithms alone omit torus holonomies and satisfy compatibility constraints; writing a global ∇_X requires additional coordinates. Gauge invariance permits the quadratic flat holonomy response (11).

A4(b), differentiation: ACCEPT with its domain hypotheses. For a fixed coordinate domain and a U -independent barrier, differentiating under the finite integral gives, for any boundary coordinates u_i ,

$$\partial_i \partial_j \mathcal{D}_s = E[S_{ij}] - \text{Cov}(S_i, S_j) + B_{ij}. \quad (12)$$

Uniform local domination justifies differentiation. A moving hard boundary needs boundary terms, unless it is first represented in a fixed domain. Equation (12) alone provides no local, independent flux coordinates with plane-uniform derivative bounds.

A4(b), covariance upper bound: REFINE. Put $V = S + W_\eta$ and let κ bound the unbarriered Hessian error on the **whole support**. The Brascamp–Lieb inequality gives

$$\text{Cov}_\mu(\xi) \leq E[(\nabla^2 V)^{-1}] \leq t(H_A - \kappa I)^{-1}. \quad (13)$$

Convex barriers preserve this upper bound. For a hard convex restriction one can obtain it by convex approximation. In (2) the available choice is $\kappa = C_3(\varepsilon + c_0)$; replacing it by $3C_3\varepsilon$ requires controlling the exterior region.

A4(b), covariance lower bound: REJECT as used. With the stated integration-by-parts hypotheses, the score identity and matrix Cauchy–Schwarz give

$$\text{Cov}_\mu(\xi) \geq (E[\nabla^2 V])^{-1}, \quad E[\nabla V \nabla V^T] = E[\nabla^2 V]. \quad (14)$$

The right-hand side contains $E[\nabla^2 W_\eta]$. Its sign increases the Fisher information and decreases this lower bound. Discarding it has the wrong direction for the asserted lower estimate. A tail probability alone bounds neither an unbounded barrier Hessian nor the Fisher information. Hard truncation produces boundary terms; already a one-dimensional centred Gaussian truncated to a finite symmetric interval has variance strictly smaller than its untruncated variance. To recover a two-sided comparison one needs, for example,

$$\|tE[\nabla^2 S] - H_A\| \leq \kappa, \quad t\|E[\nabla^2 W_\eta]\| \leq b, \quad (15)$$

in addition to the pointwise lower bound in (13). These give the conditional estimate

$$\|\text{Cov}_\mu(\xi) - tH_A^{-1}\| \leq \max \left\{ \frac{t\kappa}{4(4 - \kappa)}, \frac{t(\kappa + b)}{4(4 + \kappa + b)} \right\}. \quad (16)$$

It follows by inversion of the Loewner bounds and $H_A \geq 4I$. The original proof established neither (15) with $b = O(\varepsilon)$ nor the required interacting tail estimate. The displayed Brascamp–Lieb inequality is the standard input (Brascamp–Lieb 1976, metadata); (14)–(16) follow by the written score calculation above.

A4(b), covariant-to-abelian comparison: REJECT as justified. The kernel $M_A H_A^{-1} M_A^T$ carries adjoint transport between distinct face frames. Gauge transformations act independently at these frames; comparison with a fixed abelian matrix needs specified identifications. Only traced closed products are gauge invariant by themselves. Even for a closed product, curvature bounds control contractible loops; a winding loop can have nontrivial holonomy with every $X_p = 0$. Finally (4) controls the total operator power, whereas taking absolute values path by path loses its cancellations and introduces path multiplicity. The proposed $\sum_L L^2 3^{-L}$ argument omitted all three issues. Needed: a gauge-covariant bulk kernel comparison with summable spatial moments, and a separate winding remainder.

A4(b), nonlinear score remainder: REFINE. If local derivatives actually satisfy $E|\nabla_\xi \langle v, R \rangle|^2 \leq C(\varepsilon + \eta)^2 \|v\|^2 / t^2$ on a support with curvature error κ , then (13) implies $\text{Var} \langle v, R \rangle \leq C(\varepsilon + \eta)^2 \|v\|^2 / [t(4 - \kappa)]$. Cauchy–Schwarz bounds the cross covariance. The Taylor bound on R requires its derivative version and a controlled boundary chart; its value bound alone supplies neither.

A4(c), Taylor step: REFINE. A precise surviving implication is as follows. Suppose that winding has been separated, a genuine C^2 coordinate path $X(s)$ satisfies (10), and $h = \mathcal{D}_s - \mathcal{D}_0 - W_N$ is a function of those coordinates. Assume $X(0) = 0$, $\nabla h(0) = 0$, and along the path

$$\|\nabla^2 h\| \leq A\varepsilon/t, \quad \sup_p |\nabla_p h| \leq B\varepsilon/t. \quad (17)$$

Let q_I bound the number of p for which a fixed p' appears in $p' \sim p$. The exact chain rule and Taylor's integral formula give

$$|h(X(1)) - h(0)| \leq \frac{C_I}{2}(A + Bq_I)\frac{\varepsilon}{t} \sum_p |X_p(1)|^2. \quad (18)$$

Indeed the Hessian term integrates to at most $AC_I\varepsilon\|X(1)\|^2/(2t)$ and the gradient-acceleration term to at most $BC_Iq_I\varepsilon\|X(1)\|^2/(2t)$. An operator norm estimate alone gives an ℓ^2 gradient bound; it does not imply the per-plaquette bound in (17). A uniformly summable kernel estimate would supply that missing norm. For abelian data the linear flux path is available only in a compatible coordinate sector with the harmonic data fixed. All constants in (18) are explicit in the supplied hypotheses; proving them uniformly in the plane is still part of the problem.

6. What stays open for the weak (18) of the mid-plane note (A5 REFINE)

- (a) **Small-field comparison.** The tasks include barrier moment control, a background-covariant local response estimate, independent admissible coordinates or a replacement for the global path, and winding control. The original list of only Hypothesis I and large fields was incomplete. The formal order- t coefficients retain their status as formal coefficients; this review supplies no normalized remainder for them.
- (b) **Large fields: the failed naive Peierls arithmetic is accepted.** Using the extra premise $|Y_g| \leq \varepsilon$, the face loss at $|\xi_e| \leq 2\varepsilon$ is $\exp[-(\varepsilon + 8\varepsilon)^2/(4t)] = e^{-81\varepsilon^2/(4t)}$ per affected face, against bridge gain $e^{-4\varepsilon^2/t}$ per large edge. Since an edge touches two faces, the crude worst-case loss is $81\varepsilon^2/(2t)$ per edge and the net exponent is $+(73/2)\varepsilon^2/t$. The comparison grows rather than decays. With only the coarse small-field assumption, §4 gives $|Y_g| \leq 3\varepsilon$; replace 81 by 121 and $73/2$ by $113/2$. These are bounds on the exponential part; heat-kernel prefactors add powers of t , which cannot repair the wrong sign as $\varepsilon^2/t = t^{-2\delta} \rightarrow \infty$. Shared faces can reduce the loss for particular sets; the crude bound suffices to expose failure of the proposed uniform argument. A normalized connected-set or other large-field estimate is still required. The soft barrier also needs comparison with a genuine small-field partition function.
- (c) **Logarithmic size: conditional, with the rate corrected.** For flat Gaussian backgrounds the explicit walk count (17) in the mid-plane note proves $0 \leq \Omega_N \leq (3N/n_*)3^{-n_*}$, $n_* = \min(N_2, N_3)$. More generally, if a winding estimate

$$|W_N| \leq Ce^{-\gamma n_*} \sum_p (1 + |X_p|^2/t) \quad (19)$$

has been proved with $\gamma > 0$ independent of N, t , then $n_* \geq (\alpha/\gamma) \log(1/t) + C_0$ suffices for $P(\alpha)$, with the constant multiplied by $e^{-\gamma C_0}$. A decay factor $q = (2 + \kappa)/6$ would give $\gamma = \log(6/(2 + \kappa))$; $\log 3$ is the value at $\kappa = 0$. With fixed positive κ one must use the smaller rate. If $\kappa = O(\varepsilon)$, the $\log 3$ leading coefficient can be recovered with a bounded correction because $\varepsilon \log(1/t) \rightarrow 0$. Polynomial prefactors in n_* need absorption into a slightly smaller rate or a $\log \log(1/t)$ correction. Lemma 1 alone proves (4); a winding estimate for the normalized interacting free energy still requires a separate argument.

7. Part B: quasi-local response and the obstruction to plaquette telescoping

Result, GPT-6 Astra, 2026-09-28. A covariance version of Lemma 1 gives a plane-uniform locality theorem for local perturbations of the potential, under the explicit analytic hypotheses below. Converting this to independent plaquette perturbations needs a further gauge estimate. The natural uniformly bounded local right inverse of curl already fails at the abelian linearization, as (26) proves. Thus this step leaves Hypothesis I unreplaced; it isolates the cancellation a successful replacement must establish.

7.1 Source check and scope

Helffer–Sjöstrand (1994), *On the correlation for Kac-like models in the convex case*, J. Stat. Phys. **74**, 349–409, treats dimension-dependent Gibbs measures that are controlled perturbations of harmonic potentials (reading label: **publisher metadata and abstract**; the full journal text was behind subscription access in this check). Helffer’s 1993 author report explicitly presents their joint work (reading label: **passage**, §4, Theorem 4.2, and §5, printed pp. XII-13–17). It states exponential correlation decay for sufficiently small coupling in a nearest-neighbour harmonic model with a C^3 interaction having bounded derivatives through order three. Section 5 explains the elliptic vector equation and the additional weighted estimates needed for spatial decay. This checks a concrete source scope; the general finite-range statement needed here is proved below with its own hypotheses. The author report also addresses thermodynamic limits. No source passage checked here supplies gauge charts, barrier comparison or plaquette telescoping. Bibliography entries: `HelfferSjostrand1994`, `Helffer1993Kac`.

7.2 A covariance theorem with the same decay constants

Theorem 5 (local-potential response). Let E be the finite edge graph and $\mu \propto e^{-V} d\xi$ on $\mathbb{R}^{3E}$, with $V = S + \sum_e w_e(\xi_e)$ smooth and confining, w_e convex. Assume the usual closed weighted-gradient realization, so integration by parts and the differentiated Poisson equation hold; smooth potentials with bounded Hessians and a uniform positive lower bound suffice. Allow limits of such measures only when covariances and the gradient norms appearing below converge. Suppose, everywhere on the domain,

$$\|t\nabla^2 S - 6I\| \leq 2 + \kappa, \quad 0 \leq \kappa < 4, \quad (\nabla^2 S)_{ee'} = 0 \quad \text{if } d(e, e') > 1. \quad (20)$$

For smooth f, g with gradients supported in edge sets R, T , define $a_f = \|\nabla f\|_{L^2(\mu)}$ and $a_g = \|\nabla g\|_{L^2(\mu)}$. Then, uniformly in the graph size,

$$|\text{Cov}_\mu(f, g)| \leq \frac{t}{4 - \kappa} q^{d(R, T)} a_f a_g, \quad q = \frac{2 + \kappa}{6} < 1. \quad (21)$$

The on-site convex Hessians can be large; the bound uses their sign, without treating them as a small perturbation.

Proof. Write $L = -\Delta + \nabla V \cdot \nabla$, nonnegative in $L^2(\mu)$. Uniform convexity gives a gap at least $(4 - \kappa)/t$ for this auxiliary diffusion. Solve $Lu = g - Eg$ on the orthogonal complement of constants. Differentiating gives

$$(LI + \nabla^2 V)\nabla u = \nabla g, \quad \text{Cov}(f, g) = \langle \nabla f, (LI + \nabla^2 V)^{-1} \nabla g \rangle_{L^2(\mu)}. \quad (22)$$

The first equality is the commutator identity $\nabla Lu = (LI + \nabla^2 V)\nabla u$; the second follows by integration by parts in $\langle f - Ef, Lu \rangle$. This is the covariance representation motivating the Helffer–Sjöstrand method.

On the vector Hilbert space set

$$\mathcal{B} = tLI + 6I + t \text{diag}_e \nabla^2 w_e, \quad \mathcal{R} = t\nabla^2 S - 6I.$$

The operator $\mathcal{B}$ is block diagonal in the edge **index**, although each scalar diffusion acts on all configuration variables. It satisfies $\mathcal{B} \geq 6I$; $\mathcal{R}$ has range one and norm at most $2 + \kappa$. Therefore

$$t(LI + \nabla^2 V) = \mathcal{B} + \mathcal{R}, \quad (\mathcal{B} + \mathcal{R})^{-1} = \sum_{n \geq 0} (-\mathcal{B}^{-1} \mathcal{R})^n \mathcal{B}^{-1}.$$

Its block from T to R has zero contributions for $n < d(R, T)$. The remaining operator norm is bounded by $\frac{1}{6} \sum_{n \geq d} q^n = q^d / (4 - \kappa)$. Insert this in (22), with the factor t from the inverse scaling, to prove (21). Smooth approximation and convergence extend the inequality to the stated limits. $\square$

For the local action in §4, (20) holds on its chart with $\kappa = C_3(\varepsilon + c_0)$, independently of the background connection. Application to the singular soft-barrier measure (2) additionally needs a realization or approximation satisfying the analytic hypotheses of Theorem 5 with the same finite-range bounds. Lemma 3 is a chart estimate; constructing that approximation remains an explicit condition here. This avoids importing a whole-space theorem into a bounded chart without specifying its boundary domain. Taking $\kappa \leq 1$ gives the concrete constants $t/3$ and decay 2^{-d} as looser bounds in (21). These rates concern mid-plane fluctuations and their auxiliary diffusion.

7.3 What locality of a free-energy increment means

Suppose $V_{u,v} = V_0 + u h_R + v k_T$, $(u, v) \in [0, 1]^2$, satisfies Theorem 5 uniformly with the same t, κ . Here the gradients of h_R, k_T have supports R, T , and let $a = \sup_{u,v} \|\nabla h_R\|_{L^2(\mu_{u,v})}$, $b = \sup_{u,v} \|\nabla k_T\|_{L^2(\mu_{u,v})}$. For $F(u, v) = -\log \int e^{-V_{u,v}}$, differentiating twice yields $F_{uv} = -\text{Cov}(h_R, k_T)$. Consequently

$$|F(1, 1) - F(0, 1) - F(1, 0) + F(0, 0)| \leq \frac{tab}{4 - \kappa} q^{d(R, T)}. \quad (23)$$

Thus the increment caused in R depends exponentially weakly on a remote change in T . For a local observable f_T independent of u , the same argument gives

$$|E_1 f_T - E_0 f_T| \leq \frac{t}{4 - \kappa} q^{d(R, T)} \sup_u \|\nabla f_T\|_{L^2(\mu_u)} \sup_u \|\nabla h_R\|_{L^2(\mu_u)}. \quad (24)$$

These statements also hold for nonlinear parameter paths if the mixed potential derivative vanishes for disjoint supports, with h_R, k_T replaced by the parameter scores. Local external normalizations have zero mixed derivative for disjoint supports. In a fixed local link chart, changing coarse links in a region changes only incident bridge and face potentials, after enlarging the region by a fixed number of cells. Subject to the analytic condition after Theorem 5, (23) is therefore the precise localized-increment statement for $\mathcal{D}_s$.

Tails are summable uniformly on these two-dimensional edge graphs. A crude bound on the number of edges at graph distance ℓ from one edge is $2(4\ell + 3)^2$. Thus

$$\sum_{\ell \geq r} 2(4\ell + 3)^2 q^\ell \leq C(q) q^{r/2}, \quad C(q) = 2 \sum_{\ell \geq 0} (4\ell + 3)^2 q^{\ell/2} < \infty. \quad (25)$$

Summing (23) over remote local scores with bounded gradient norms gives this tail, with their support-size and support-radius constants made explicit in a, b and the distance. The unlocalized increment itself may have size proportional to the changed region; (23) controls its dependence on the distant environment.

7.4 The estimate that fails in the proposed telescoping

Plaquette variables obey compatibility relations. A single plaquette update at fixed surrounding data generally fails those relations. Even in the linearized embedded $U(1)$ sector on an $n \times n$

torus, with harmonic data fixed, infinitesimal link changes a and flux changes x satisfy $Ca = x$ and $\sum_p x_p = 0$. At momentum $k = (2\pi/n, 0)$ the curl symbol of the mid-plane note is $c(k) = (0, e^{2\pi i/n} - 1)$. For a flux mode supported at this momentum, every solution satisfies

$$\|a\|_2 \geq \frac{\|x\|_2}{2 \sin(\pi/n)} \geq \frac{n}{2\pi} \|x\|_2. \quad (26)$$

Real sine/cosine modes give the same bound. Hence a right inverse $a = \mathcal{R}x$ on admissible zero-mean fluxes cannot have norm bounded independently of n . An exponentially local kernel $\|\mathcal{R}_{ep}\| \leq Ce^{-\gamma d(e,p)}$ with C, γ independent of n would have bounded row and column sums by (25), and bounded ℓ^2 norm by the Schur test, contradicting (26). Changing two separated opposite plaquette fluxes can instead use a long string of changed links or a spatially spread representative. The supports and score norms in (23) then track that representative.

The precise missing step is a **curvature response estimate after gauge cancellations**, rather than a local right inverse. For example, in specified admissible flux coordinates with harmonic data retained and winding subtracted, one would need the normalized remainder $h = \mathcal{D}_s - \mathcal{D}_0 - W_N$ to have a Hessian kernel K_{pq} satisfying

$$\sup_p \sum_q e^{\gamma d(p,q)} \|K_{pq}\| \leq C\varepsilon/t, \quad (27)$$

together with compatible paths or local replacements whose total quadratic cost is bounded by $C \sum_p |X_p|^2$, and the required zero-flux/winding remainder. Equation (21) controls the link-score covariance. The cancellation of the large inverse-curl factors in the combined expression $E[S_{ij}] - \text{Cov}(S_i, S_j) + B_{ij}$ remains to be proved. The Gaussian case exhibits such cancellations; (26) therefore diagnoses failure of the naive coordinate conversion, without ruling out (27).

With those extra estimates, (18), or a compatible telescoping version of it, and (19) would give the weak $P(\alpha)$ bound on the selected large-plane regime. Theorem 5 and locality alone supply neither the $O(\varepsilon)$ comparison to the Gaussian response nor the winding bound, and Part A's barrier and large-field issues remain. This is the precise obstruction reached in Part B. All claims of $P(\alpha)$ and the normalized expansion (18) of the mid-plane note remain conditional.

8. Round 9: cancellation in flux response

After refereeing (Claude, 2026-09-28). §8.1 ACCEPT, checked in full: Woodbury gives $G = \frac{1}{4}(I - C^*PC)$, hence (28); the slab form (29) vanishes at constant flux and gives $-|x|^2/(4t)$ for opposite layers, both as in Proposition 2; $8I + L = 12I - A$ and the walk series give the weighted row sum $1/(12 - 4e^\gamma) = \frac{1}{6}$ at $\gamma = \log(3/2)$, so (30) reads $\frac{1}{t}(\frac{5}{12} + \frac{1}{4}) = \frac{2}{3t}$. §8.2 ACCEPT as a formal statement: the equal-layer symbol $M(p) = -\lambda/(2(8 + \lambda))$ follows from (29), a flat neutral connection shifts it to $M(p + H)$, and the Coulomb lift $X^3(k)/(e^{i\kappa} - 1)$ of a soft flux produces the $1/\kappa$ residue in (32). The pole is a property of the fixed-frame chart (parallel transport of charged curvature), consistent with the gauge-invariant expectation of §6; it fixes that (27) must be posed covariantly. §8.3 ACCEPT.

8.1 Exact Gaussian reference (GPT-6 Astra, 2026-09-28)

Use the unrestricted abelian Gaussian in one slab with boundary links a_0, a_1 . Retain their harmonic components h_0, h_1 and impose the linear compatibility relations on all cut and transverse fluxes. Put $x = Ca_0$, $y = Ca_1$, so $\sum x = \sum y = 0$. Independent zero-mean Fourier components and harmonic data give actual coordinates. The spatial kernels below represent bilinear forms on this constrained space, by restriction of an ambient kernel; eliminating one face or inserting a nonlocal orthogonal projector would change its row sums. The three colour copies in the linearization are identical. Set $L = CC^*$, $P = (8I + L)^{-1}$ and $G = H_0^{-1}$. The mean bridge

link is $b = (a_0 + a_1)/2$ up to a gradient from direction-1 links. In uncentred mid-links m the normalized exponent and external old-face term are

$$S_0 = \frac{2}{t} \|m - b\|^2 + \frac{1}{4t} \|Cm\|^2, \quad B_0 = -\frac{\|x\|^2 + \|y\|^2}{8t}.$$

Assign half of each old-face ratio to each slab. Since $\text{Cov}_0(m) = tG$, differentiation in b gives

$$E_0[S_{0,bb}] - \text{Cov}_0(S_{0,b}, S_{0,b}) = \frac{1}{t}(4I - 16G) = \frac{4}{t}C^*PC, \quad G = \frac{1}{4}(I - C^*PC). \quad (28)$$

For any lift $\mathcal{R}$ with $C\mathcal{R} = I$ on admissible fluxes, $(C\mathcal{R})^*P(C\mathcal{R}) = P$ as a restricted bilinear form. The large factors in (26) disappear. Gaussian integration consequently gives

$$\begin{aligned} \mathcal{D}_{0,\text{slab}} &= \frac{1}{2t} \langle x + y, P(x + y) \rangle - \frac{\|x\|^2 + \|y\|^2}{8t} + \text{const}, \\ K_{\sigma p, \tau q}^0 &= \frac{1}{t} \left(P_{pq} - \frac{1}{4} \delta_{\sigma\tau} \delta_{pq} \right) I_3, \quad \sigma, \tau \in \{0, 1\}. \end{aligned} \quad (29)$$

Cut-flux directions at fixed x, y and all harmonic directions have zero response. Summing slabs reproduces Proposition 2 of the series/parallel note (passage). On the square torus $8I + L = 12I - A$, where A is nearest-neighbour adjacency. Its positive walk series yields $\sup_p \sum_q e^{\gamma d(p,q)} P_{pq} \leq (12 - 4e^\gamma)^{-1}$. For $\gamma = \log(3/2)$ and the pair distance $d((\sigma, p), (\tau, q)) = d(p, q) + |\sigma - \tau|$, (29) therefore gives

$$\sup_{\sigma, p} \sum_{\tau, q} e^{\gamma d} \|K_{\sigma p, \tau q}^0\| \leq \frac{1}{t} \left(\frac{1 + e^\gamma}{12 - 4e^\gamma} + \frac{1}{4} \right) = \frac{2}{3t}. \quad (30)$$

Assembly gives $4/(3t)$, independently of the periods. The Gaussian remainder h has zero kernel, hence satisfies (27) with $C = 0$; (30) bounds the reference response itself.

Mechanism. The Gaussian score identity produces the Schur complement (28), equivalently the Woodbury identity for $4I + C^*C/2$. Its two exterior curls annihilate gradients and harmonic shifts and turn every flux lift into the identity. Combining the direct Hessian and covariance before taking absolute values leaves a massive face resolvent. Normalized bridge denominators remove the cut-face response; the old-face normalization adds only the diagonal term in (29). This calculation uses the full Gaussian domain; a barrier requires its own correction.

8.2 Formal non-abelian response: a surviving soft transport term

Specify a chart before testing (27). Use face logarithms in fixed site frames, with the formal Coulomb lift $a = \mathcal{R}X + H + O(X^2)$, $\mathcal{R} = C^*L^{-1}$ on zero-mean modes; retain H and all nonlinear compatibility relations. This is a finite-volume formal chart about zero. Its cubic response already contains an inverse-curl pole. The following test concerns equal adjacent layers, zero cut flux, and $H = 0$.

Use the unrestricted zero-image expansion; barrier and image errors remain outside it. Write the normalized action as $S_0 + V_3 + V_4 + J + \dots$, with $V_j = \mathcal{S}_j/t$, and expand the BCH logarithm as $Z_1 + Z_2 + Z_3 + \dots$. Each squared mid-face contributes $\langle Z_1, Z_2 \rangle / (2t)$ and $(|Z_2|^2 + 2\langle Z_1, Z_3 \rangle) / (4t)$ to V_3, V_4 respectively. The bridge curvature contributes $-\sum_e [X_e, \xi_e]^2 / (24t)$ to V_4 ; J includes the normalized Haar and heat-kernel amplitudes. With E_0 at the shifted Gaussian saddle, the first insertions and the required connected cubic pair are

$$\Delta \mathcal{D} = E_0 V_3 + E_0 V_4 - \frac{1}{2} \text{Cov}_0(V_3, V_3) + E_0 J + \dots \quad (31)$$

One vertex is first order in an insertion parameter; under $[\ , \] \mapsto g[\ , \]$, the cubic is order g , while quartic and the connected pair are both order g^2 . Gaussian Wick contraction gives the classical terms $F_3/t, F_4/t$ in (15) of the order- t note (passage), including $-\langle \partial_\xi \mathcal{S}_3, G \partial_\xi \mathcal{S}_3 \rangle / (2t)$.

Thus a one-vertex quartic truncation alone would miss a necessary term. The complete classical cubic can be tested without listing its vertices. For equal layers, (29) has flux Hessian $M(p)/t$, where $M(p) = -\lambda(p)/(2(8 + \lambda(p)))$ and $\lambda(p) = 4 - 2\cos p_2 - 2\cos p_3$. At a constant flat neutral connection $a_j = H_j T_3$, the charged components have the exact classical quadratic symbol $M(p + H)/t$. Indeed replace C by the flat covariant curl in (28); its face Laplacian has symbol $\lambda(p + H)$. This also fixes the sum of **all** cubic transport vertices at zero incoming connection momentum. Massive Gaussian elimination makes these link vertices analytic near that momentum. For a neutral soft flux at $k = (\kappa, 0)$, the Coulomb lift has $a_3^3(k) = X^3(k)/(e^{i\kappa} - 1)$ and $a_2^3(k) = 0$; here $X(z) = \sum_k X(k)e^{ikz}$ defines the Fourier amplitudes. The resulting first-order charged flux Hessian of h , with outgoing momenta $p + k, p$, consequently contains

$$\begin{aligned} K_+^{(1)}(p + k, p) &= \frac{X^3(k)}{t} \left[\frac{\partial_{p_3} M(p)}{e^{i\kappa} - 1} + O(1) \right], \\ \partial_{p_3} M(p) &= -\frac{8 \sin p_3}{(8 + \lambda(p))^2}. \end{aligned} \tag{32}$$

Choose $p_3 \notin \{0, \pi\}$ and fixed nonzero p . The bounded term includes the other placements of the neutral leg; their inverse curls have momenta near p , and remain bounded. Reality pairs k with $-k$. Equation (32) has size $|X^3(k)|/(t|\kappa|)$ with a nonzero residue. A plane-uniform row-sum bound $C \sup |X|/t$ would bound every such Fourier matrix element and is incompatible with this coefficient as $\kappa \rightarrow 0$. This is a formal failure of (27) in this fixed-frame chart. It occurs in the classical cubic, before quartic terms: the latter and the cubic pair have classical boundary degree four, and bridge curvature vanishes on this equal-layer test. Loop contractions have higher powers of t than the displayed $1/t$ residue. Winding subtraction removes wrapping paths; the nonwrapping massive resolvent has the same bulk momentum shift and residue. A flat-holonomy subtraction cannot remove it.

The mechanism is parallel transport of charged curvature between distinct faces. The Gaussian Schur complement cancels the inverse curls on its two external legs; differentiating its transport introduces a third leg $\mathcal{R}X$, retained by (32). Covariantly transported derivatives or another specified flux chart could change this conclusion. Such a replacement needs its own definition and bound; (32) identifies the term that prevents promoting the Gaussian proof in fixed site frames.

8.3 Beyond first order

To pursue (27), first replace the fixed-frame Hessian by a specified covariant response that absorbs (32), and prove the corresponding compatible-path estimate. Then control all connected insertions with plane-uniform weighted sums, the normalized interacting barrier moments, the full winding remainder and the large-field comparison. The Gaussian identity and formal expansion supply no such all-order remainder bound.

9. Consequence for STATE

Atlas cell 2 keeps its formal order- t calculation. Round 8 Part A accepts the covariant decay and corrected bridge/convexity bounds, rejects Proposition 4 as stated, and replaces its Taylor step by the explicit conditional implication (17)–(18). The next small-field task is the normalized, gauge-covariant response estimate with barrier and winding control. Part B proves locality for local-potential increments under Theorem 5's analytic hypotheses and isolates (27), including its admissible-coordinate and cancellation requirements, as the unresolved step toward replacing Hypothesis I. Round 9 §8.1 establishes the exact Gaussian cancellation and its plane-uniform flux-kernel bound; §8.2 finds a formal cubic transport obstruction in the fixed-frame Coulomb chart. A covariant replacement for (27) is the next response task. The large-field and iteration obligations remain.